

Heitor Marques Simões Barbosa

Ph.D. Student in Computer Science, Tulane University

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Education

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| Ph.D. Tulane University , Computer Science
• Advisor: Dr. Hridesh Rajan, Laboratory for Software Design. | New Orleans, LA, USA
Aug 2026 – present |
| B.S. University of Brasília (UnB) , Software Engineering | Brasília, Brazil
June 2020 – July 2025 |

Publications

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| DataFabric-DermAlert: A Tool for Federated Health Data Integration and Fairness-Aware Dataset Construction
Abdul Hannan, Heitor Marques S. Barbosa , Carla Silva Rocha Aguiar, Isaque Alves
(Brazilian Symposium on Software Engineering (SBES 2026), São Paulo, Brazil) | Sept 2026 |
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Research Experience

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| DermAlert Project, University of Brasília , Researcher
• Co-developed DataFabric-DermAlert, an open-source tool for federated health data integration, semantic equivalence management, and versioned, auditable dataset construction (SBES 2026).
• Contributed to a data fabric architecture for capturing, storing, and analyzing dermatology images in Brazil's Basic Health Units (UBS) to support early skin cancer screening. | Brasília, Brazil
Jan 2025 – July 2026 |
| PAULA Project, University of Brasília , AI Team Lead
• Led the AI team of PAULA (Paranoá Alfabetizando Usando Letramento Analógico), a UnB extension project that develops an offline mobile app, based on Paulo Freire's method, to help semi-literate and illiterate residents of Paranoá learn to read and write.
• Developed handwriting-recognition models (convolutional and transformer-based networks, stroke analysis) that read and correct letters written with the finger on the phone screen.
• Designed backend APIs to serve models and process data efficiently and securely. | Brasília, Brazil
Mar 2023 – Dec 2023 |
| AI.lab, University of Brasília , Machine Learning Researcher
• Worked on the artificial intelligence track of Justiça 4.0 (CNJ/UNDP), a program for the digital transformation of the Brazilian Judiciary, under the coordination of Dr. Nilton Correia da Silva and Dr. Fabrício Ataídes Braz.
• Implemented and evaluated clustering algorithms (K-means, GMM, spectral clustering, Kohonen networks, DBSCAN) to group similar court cases and discover topics for intelligent triage.
• Built text-preprocessing pipelines versioned with DVC and tracked experiments with MLflow; used multiprocessing to scale to large document collections.
• Collaborated with legal experts to validate labels, improving model accuracy and supporting the modernization of the judiciary. | Brasília, Brazil
July 2022 – Dec 2023 |

Professional Experience

Forlex, AI Engineer

Remote

July 2025 – July 2026

- Designed and evaluated retrieval-augmented generation (RAG) pipelines for legal documents, benchmarking embedding models and chunking strategies to maximize context retrieval.
- Fine-tuned small language models (SLMs) for high-precision legal tasks, with a focus on data consistency and model performance.

SantoDigital, Machine Learning Specialist

Remote

Jan 2025 – July 2025

- Developed and trained computer vision and AI models for document processing (SantoID).
- Deployed models on Google Cloud Platform (Cloud Storage, Vertex AI, Cloud Run) and built API endpoints for integration with external systems.

Poder360, Data Engineer

Remote

Feb 2023 – Jan 2025

- Built a Celery-based automation system with scrapers that collect, process, and analyze large volumes of data from news agencies and government bodies in real time.
- Adopted MongoDB with indexing and query optimization, and deployed the system to production, enabling faster monitoring of emerging trends.

Brazilian Ministry of Education (MEC), Intern

Brasília, Brazil

Dec 2021 – June 2022

- Implemented computer vision models that recognize fields in historical medical residency record books and extract their handwritten content.
- Built business intelligence dashboards on medical and multidisciplinary residency programs, and designed an information security plan for sensitive data.

Volunteer

Debian Project, Contributor

Remote

Aug 2023 – Mar 2024

- Packaged and maintained software for the Debian stable and unstable branches in compliance with the Debian Policy.
- Automated build processes with CI/CD pipelines and debugged low-level compilation errors.

Technical Skills

Programming: Python, C/C++

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, spaCy, TRL, ms-swift

MLOps: MLflow, DVC, TensorBoard, model monitoring

Data and Cloud: PostgreSQL, MongoDB, MySQL, AWS, Google Cloud Platform

Software: FastAPI, Django, Docker, Kubernetes, CI/CD